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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION October 2025
Third SEMESTER
Discipline Specific Core Course (DSC) - MG3DSCEVS200
NATURAL RESOURCES AND CONSERVATION
(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate (E), Create (C), Skill (S), Interest (1) and Appreciation (Ap)
Students should attempt at least one question from each course outcome to enhance their overall outcome attainability

Part A
Short answer - Answer any 10 questions - Each question carries 1 mark

1. **What are the uses of coal?** **(U)/ (CO 1)**
   Coal is used as a fuel for cooking and heating/ in thermal power plants to generate electricity /in industries such as iron and steel production/ to make by-products like coke, coal tar, and coal gas (Any two)
2. **Give an example of a localized abiotic resource.** **(K)/ (CO 1)**
   Minerals like mica/ Limestone deposits/ Natural gas Marble/water/rock/land/soil /air (Any one)
3. **Define animal resources** **(K)/ (CO 5)**
   Animal resources are animals and their products used by humans for food, clothing, work, or other purposes. (Examples: milk, meat, wool, leather, honey, etc.)
4. **Give an example of a microbe used as a bio fertilizer.** **K)/ (CO 5)**
   Rhizobium/ Azotobacter/ Azospirillum/ Cyanobacteria/ Mycorrhiza/ Phosphate solubilizing bacteria (PSB)
5. **Explain the importance of water management** **(K)/ (CO 2)**
   Water management is important to ensure the proper use, distribution, and conservation of water resources. It helps prevent water scarcity, supports agriculture, maintains ecosystems, ensures safe drinking water, and reduces the effects of drought and floods.
6. **Define water scarcity** **(K)/ (CO 2)**
   Water scarcity is a situation where the demand for water is greater than the available supply of water in a region.
7. **Describe the importance of water conservation.** **(K)/ (CO 2)**
   Water conservation helps to save water for future generations, reduces water pollution, protects natural habitats, ensures sustainable agriculture, and helps deal with climate change and drought conditions.
8. **How does climate affect the distribution of freshwater resources?** **(U)/ (CO 2)**
   Regions with high rainfall have abundant water, while dry or hot areas face water shortage. Climate change can also cause uneven distribution through droughts, floods, and melting glaciers.
9. **What is the importance of sustainable land management practices?** **(U)/ (CO 3)**

Sustainable land management helps to protect soil fertility, prevent land degradation, support agriculture, conserve biodiversity, reduce pollution, and ensure that land resources are available for future generations.

10. **Illustrate the threats due to deforestation (U)/ (CO 4)**

Loss of biodiversity/ Soil erosion and loss of fertile land/ Increase in carbon dioxide causing global warming/Disturbance of the water cycle and rainfall pattern/Increased risk of floods and droughts.

11. **Identify the importance of forest resources. (U)/ (CO 4)**

Timber, fuelwood, medicinal plants, and raw materials/Habitat for wildlife and biodiversity/Oxygen release and carbon dioxide absorption/ Prevention of soil erosion and control of floods / Livelihood to tribal and rural communities

12. **Give any two examples for social aspects of forest (K)/ (CO 4)**

Livelihood support/Cultural importance / Recreation and tourism/ Traditional knowledge & medicine / medicines /Shelter and settlement

(1 x 10 = 10)

Part B

Short Essay - Answer any 6 questions - Each question carries 5 marks

13. **What are the main advantages and disadvantages of solar energy? (K)/ (CO 1)**

- Renewable and inexhaustible source of energy
- Eco-friendly, produces no pollution
- Reduces electricity bills
- Suitable for remote and rural areas
- Low maintenance cost after installation

Disadvantages:

- High initial installation cost
- Depends on sunlight, so not efficient at night or in cloudy weather
- Requires large area for solar panels
- Energy storage (batteries) is expensive

14. **How does over exploitation of natural resources affect biodiversity? (U)/ (CO 1)**

- Habitat destruction – animals and plants lose their living space
- Extinction of species due to overhunting, overfishing, etc.
- Loss of genetic diversity in plants and animals
- Disturbance of food chains and ecosystems
- Reduced population of endangered species

15. **Discuss the importance of ex-situ conservation for critically endangered species (U) / (CO 5)**

Ex-situ conservation means protecting endangered species outside their natural habitat.

Importance

- Protects species from threats like habitat loss, poaching, climate change
- Helps in breeding programs to increase population
- Allows scientific study and genetic preservation (gene banks, seed banks)
- Can reintroduce species into the wild when conditions improve
- Examples: Zoos, botanical gardens, seed banks, cryopreservation.

16. **Describe the importance of wetland resources with examples. (U)/ (CO 2)**

Wetlands are areas like marshes, swamps, lakes, and estuaries.

Importance:

- Natural water filters that purify water
- Control floods by storing excess water

- Rich in biodiversity (fish, birds, plants, etc.)
- Support fishing, agriculture, and tourism
- Act as carbon sinks, reducing global warming

Examples:

Chilika Lake (Odisha) – supports fisheries and migratory birds
Keoladeo National Park (Rajasthan) – famous bird sanctuary
Vembanad Lake (Kerala) – supports agriculture and fishing

17. Write a short note on eutrophication (K)/ (CO 2)

Eutrophication is the excessive enrichment of water bodies with nutrients, mainly nitrates and phosphates, usually from agricultural runoff, sewage, and industrial waste. This causes rapid growth of algae, known as algal bloom, which blocks sunlight from reaching aquatic plants. When algae die and decompose, the oxygen in water decreases, leading to oxygen depletion (hypoxia). As a result, fish and other aquatic life die, causing loss of biodiversity. Eutrophication also leads to foul smell, bad taste in water, and makes the aquatic environment unsuitable for drinking and recreation. It is a major threat to lakes, rivers, ponds, and coastal waters.

18. Explain the importance of soil conservation techniques in detail. (U)/ (CO 3)

Importance and Techniques

- Contour ploughing – Ploughing across slopes reduces soil erosion by slowing down water flow.
- Terracing – Flat terraces are made on steep slopes to prevent soil loss in hilly areas.
- Afforestation – Planting trees helps bind soil with roots and reduces erosion by wind and water.
- Mulching – Covering soil with straw or leaves reduces evaporation and maintains soil moisture.
- Crop rotation – Growing different crops in sequence maintains soil nutrients and reduces pests.

19. Classify the different strategies for reforestation. (U)/ (CO 4)

Reforestation is the process of replanting trees in deforested or degraded areas. Strategies can be classified as:

1. **Natural Reforestation** - Forests are allowed to regrow naturally without human interference. Seeds germinate from soil seed banks or nearby trees.
2. **Artificial Reforestation** -Human-assisted planting of trees.

Methods include

- Direct seeding – sowing seeds directly into soil
- Planting seedlings or saplings raised in nurseries
- Aerial seeding – dropping seeds from aircraft over large areas

Agroforestry

- Growing trees along with crops or livestock to improve soil fertility and income.
- Social forestry Tree planting by community or local people for meeting fuel, fodder, and timber needs

20. Explain the role of sustainable forest management practices. (U)/ (CO 4)

Sustainable forest management (SFM) ensures that forests are used in a way that meets present needs without harming future generations.

Roles and Benefits:

- Maintains biodiversity – Protects habitats and prevents species extinction.
- Controls deforestation – Promotes regulated cutting and replanting of trees.
- Supports local communities – Provides livelihood, food, medicine, and raw materials.
- Prevents soil erosion and climate change – Trees absorb CO_2 and protect soil.

- Encourages eco-friendly industries – Promotes bamboo, herbal, and non-timber forest products.
- Certification programs – FSC (Forest Stewardship Council) labels ensure wood is harvested responsibly.

SFM integrates economic, social, and environmental values, making forests long-lasting and productive.

[5 x 6 = 30]

Part C

Essay - Answer any 1 questions - Each question carries 10 marks

21. Explain the major threats to mangrove resources and describe the importance of their conservation. (U)/ (CO 5)

Major Threats to Mangroves

- Mangroves are cleared for agriculture, aquaculture (especially shrimp farming), and urban development.
- Industrial chemicals, oil spills, plastic waste, and sewage destroy mangrove ecosystems.
- Rising sea levels and increased cyclones damage mangrove habitats.
- Excessive cutting of mangrove wood for fuel, timber, and charcoal weakens the ecosystem.
- Removal of mangroves makes shorelines more prone to erosion and flooding.

Importance of Conservation

- Mangroves act as natural barriers against storms, cyclones, and tsunamis.
- They provide habitat for fish, crabs, birds, and endangered species.
- Many commercially important fish breeds in mangroves.
- Mangroves absorb large amounts of carbon, helping reduce global warming.
- Mangrove roots stabilize soil and prevent coastal land loss.

22. Compare the causes of soil degradation and soil erosion. Describe its impacts. (U)/ (CO 3)

1. Soil degradation is the decline in soil quality due to physical, chemical, or biological changes

   Main causes

   Overuse of fertilizers, Industrial pollution- Salinization & alkalinization, Loss of organic matter

   High chemical use, mining, waste dumping resulting in loss of fertility and soil health

   Human Influence- deforestation and poor agriculture
2. Soil erosion Removal of topsoil by wind, water, or human

   Main causes

   Overgrazing, Heavy rainfall & wind resulting loss of top fertile layer, leading to barren land

   Impacts of Soil Degradation and Erosion

   - Loss of soil fertility – Reduces agricultural productivity and food security.
   - Desertification – Land becomes dry and unproductive.
   - Water pollution – Eroded soil sediments block rivers, lakes, and dams.
   - Loss of biodiversity – Soil organisms like earthworms and microbes decline.
   - Economic loss – Farmers suffer reduced crop yield and income.
   - Climate effect – Poor soil stores less carbon, increasing greenhouse gases.

(1 x 10 = 10)